

## Introduction

Before running a two-group comparison, the investigator needs to know how many observations per group will be required to detect a difference of scientific importance with acceptable probability. Power analysis for the two-sample t-test answers this question by linking four quantities: the true effect size, the significance level, the power, and the sample size. Given any three, R can solve for the fourth.

## Prerequisites

Familiarity with the two-sample t-test and with the concepts of type I and type II error is assumed. The reader should know that “power” means  $1 - \beta$ , the probability of rejecting  $H_0$  when  $H_1$  is true.

## Theory

Under  $H_0$ , the t statistic follows a central t distribution with  $\nu = n_1 + n_2 - 2$  degrees of freedom (Student’s test, equal variances). Under  $H_1$  where  $\mu_1 - \mu_2 = \delta$ , the statistic follows a *non-central* t distribution with non-centrality parameter

$$\lambda = \frac{\delta}{\sigma \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}.$$

Power is the probability that the test statistic exceeds the critical value under this non-central distribution. For Cohen’s standardised effect size  $d = \delta/\sigma$  and equal group sizes  $n_1 = n_2 = n$ , this simplifies to  $\lambda = d\sqrt{n/2}$ .

The four power quantities and their trade-offs:

- **Effect size**  $d$ : the minimum standardised difference considered scientifically meaningful. By Cohen’s conventions,  $d = 0.2$  is small,  $0.5$  is medium,  $0.8$  is large – but these conventions are field-dependent and should be used with judgement.
- **Significance level**  $\alpha$ : usually  $0.05$  for two-sided testing.
- **Power**: conventionally  $0.80$  for exploratory work,  $0.90$  for confirmatory or registered trials.
- **Sample size per group**  $n$ : what we typically solve for.

## Assumptions

Power analysis inherits the assumptions of the t-test it plans for: independent groups, approximately normal data (or large enough  $n$  for the CLT), and equal variances for Student’s test (unequal for Welch’s).

## R Implementation

```
library(pwr)

pwr.t.test(d = 0.5, sig.level = 0.05, power = 0.80,
           type = "two.sample", alternative = "two.sided")

pwr.t.test(n = 30, sig.level = 0.05, power = 0.80,
           type = "two.sample", alternative = "two.sided")

pwr.t.test(n = 30, d = 0.5, sig.level = 0.05,
           type = "two.sample", alternative = "two.sided")

library(pwrss)
pwrss.t.2means(mu1 = 110, mu2 = 100, sd1 = 18, sd2 = 15,
               power = 0.80, alpha = 0.05)
```

The first `pwr.t.test()` call asks: how many subjects per group do I need to detect a medium-sized effect ( $d = 0.5$ ) with 80% power? The second asks: given  $n = 30$  per group, what is the minimum detectable effect at 80% power? The third asks: given  $n = 30$  and  $d = 0.5$ , what is my power? The `pwrss:pwrss.t.2means()` call works directly in raw-mean terms, which is often more natural when planning a clinical trial.

## Output & Results

The first call returns  $n \approx 64$  per group – 128 subjects in total. The second returns a minimum detectable  $d \approx 0.735$ , corresponding to a raw difference of about 11 points if  $\sigma = 15$ . The third returns a power of approximately 0.48, showing that 30 per group is badly underpowered for a medium effect.

```
Two-sample t test power calculation

      n = 63.76576
      d = 0.5
sig.level = 0.05
  power = 0.8
alternative = two.sided
```

## Interpretation

The recommended manuscript phrasing is: “Assuming a between-group standardised mean difference of  $d = 0.5$  (a medium effect by Cohen’s criteria), a two-sided  $\alpha$  of 0.05, and power of 0.80, the required sample size is 64 per group. Allowing for 10% loss to follow-up, we aim to enrol 72 per group (144 total).” The power analysis should be reported in the methods section of the grant or protocol, not buried in an appendix.

## Practical Tips

- Base your effect size on published data or a pilot, not on a convention like  $d = 0.5$ . If the published effect is  $d = 0.3$ , planning for  $d = 0.5$  will leave you underpowered.
- Use Welch’s sample size for unequal variances (`pwr.t2n.test()` or `pwrss.t.2means()` with `sd1 != sd2`). It is slightly more conservative, which is what you want at the planning stage.
- Plan for loss to follow-up, noncompliance, and data exclusion by inflating the computed  $n$  by a realistic percentage (typically 10-25%).
- If the minimum clinically important difference (MCID) is known in raw units, use that directly rather than translating to  $d$ . It is more interpretable to clinicians.
- Always produce a sensitivity analysis: a curve of power against  $n$ , or a tornado plot of sample size against plausible values of effect size and SD.

## For Reviewers

What to look for in a paper using this method.

- Common misapplications.
- Diagnostics that should be reported but often aren’t.
- Red flags in tables and figures.
- What to verify.
- What an adequate Methods paragraph must contain.

## See also — labs in this chapter

- Sample size, power, and Quarto reporting
- Power: closed-form and simulation